



## WEEK 4 – DATA MODEL DESIGN

### End-to-End Salesforce CRM Transformation for Customer Operations and Service Management

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#### 4.1 Object Definitions

To establish a scalable and maintainable Salesforce architecture while avoiding data redundancy, a combination of standard Salesforce objects and custom objects has been designed. This structure accurately represents customer information, service activities, and utility infrastructure assets within the organization.

##### 4.1.1 Standard Objects

###### **Account**

The Account object represents households or customers receiving utility services from GreenGrid Utilities. It serves as the primary entity for customer-related information.

###### **Contact**

The Contact object stores personal details associated with customers, including names, phone numbers, and communication-related information.

###### **Case**

The Case object functions as the central record used to capture, manage, prioritize, and resolve customer service requests and reported issues.

##### 4.1.2 Custom Objects

###### **Utility\_Asset\_\_c**

This custom object represents physical utility infrastructure, specifically utility meter equipment installed at customer locations. It contains the meter identification information required to support efficient customer searches and asset tracking.

## Field\_Visit\_\_c

This custom object records field service activities, including technician assignments, maintenance visits, repair work, travel details, and resolution outcomes completed on-site.

## 4.2 Field Definitions

The following fields support the required business processes, validation controls, and performance objectives, including rapid customer search functionality.

### Custom Object: Utility\_Asset\_\_c

| Field Label          | API Name                | Data Type        | Description and Validation Rules                                                                      |
|----------------------|-------------------------|------------------|-------------------------------------------------------------------------------------------------------|
| Asset Name           | Name                    | Auto Number      | System-generated identifier using the format MET-{0000}.                                              |
| Meter ID             | Meter_ID__c             | Text (50)        | Unique indexed identifier used for fast customer searches. Configured as both Unique and External ID. |
| Asset Status         | Asset_Status__c         | Picklist         | Indicates the operational condition of the asset. Values include Active, Inactive, and Suspended.     |
| Installation Address | Installation_Address__c | Text Area (255)  | Stores the physical installation location of the utility asset.                                       |
| Customer Account     | Customer_Account__c     | Lookup (Account) | Links the utility asset to its associated customer account.                                           |

### Custom Object: Field\_Visit\_\_c

| Field Label          | API Name                | Data Type               | Description and Validation Rules                                                                                              |
|----------------------|-------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Visit Number         | Name                    | Auto Number             | Automatically generated record number using the format VIS-{0000}.                                                            |
| Parent Case          | Parent_Case__c          | Lookup (Case)           | Connects the field visit to its corresponding support case. Deletion controls are applied to preserve historical information. |
| Repair Status        | Repair_Status__c        | Picklist                | Required field used to monitor work progress. Available values: Assigned, In Progress, Resolved, Failed.                      |
| Technician Notes     | Technician_Notes__c     | Long Text Area (32,000) | Stores technician observations, findings, and repair details entered through mobile devices.                                  |
| Resolution Timestamp | Resolution_Timestamp__c | Date/Time               | Automatically populated when a repair status changes to Resolved, supporting KPI and performance reporting.                   |

## 4.3 Relationship Mapping

This section describes how standard and custom Salesforce objects are connected within the GreenGrid Utilities data model. Proper relationship design ensures data consistency, maintains historical records, and supports long-term compliance requirements.

| Parent Object    | Child Object     | Relationship Type | Data Behavior                                                                                           |
|------------------|------------------|-------------------|---------------------------------------------------------------------------------------------------------|
| Account          | Contact          | Standard Lookup   | If an Account is removed, related Contacts remain while the Account reference is cleared.               |
| Account          | Utility_Asset__c | Custom Lookup     | Asset records remain intact if customer ownership changes and can be reassigned to a different Account. |
| Account          | Case             | Standard Lookup   | Account deletion is restricted when associated Cases exist, protecting historical support records.      |
| Utility_Asset__c | Case             | Custom Lookup     | Cases remain preserved even if utility assets are replaced or updated.                                  |
| Case             | Field_Visit__c   | Custom Lookup     | Active Field Visit records containing unresolved work details cannot be deleted.                        |

## Relationship Design Explanation

### Account → Contact

A Lookup relationship is used to associate multiple Contact records with a single customer Account. This allows households or organizations to maintain several related contacts while preserving independent records.

### Account → Utility\_Asset\_\_c

A Lookup relationship connects customer accounts with utility assets. This design separates physical infrastructure from customer ownership, allowing utility assets to remain associated with a location even when ownership changes.

### **Account → Case**

A Lookup relationship with restricted deletion ensures that historical support interactions remain available for reporting, auditing, and service-level monitoring purposes.

### **Utility\_Asset\_\_c → Case**

This relationship links customer service cases directly to the utility asset involved. Support agents can quickly review maintenance history and previous issues to improve troubleshooting and resolution efficiency.

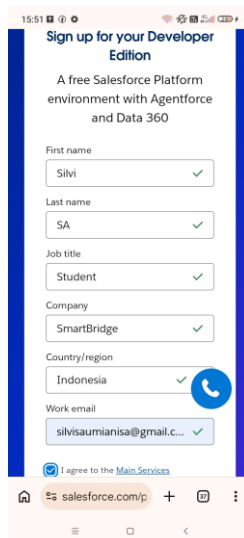
### **Case → Field\_Visit\_\_c**

A Master-Detail relationship ensures that field service activities always belong to an authorized Case record. Child records inherit ownership, visibility, and sharing settings from the parent Case, maintaining consistent security controls.

## **4.4 Salesforce Implementation and Configuration**

### **Step 1: Create a Salesforce Developer Account**

1. Navigate to the Salesforce Developer Edition registration page.
2. Complete the registration form with the required information.
3. Submit the form and wait for the account activation email.



The screenshot shows a mobile browser interface for the Salesforce Developer Edition registration page. The page title is "Sign up for your Developer Edition". Below the title, it says "A free Salesforce Platform environment with Agentforce and Data 360". The form contains several input fields, each with a green checkmark indicating successful validation: "First name" (Silvi), "Last name" (SA), "Job title" (Student), "Company" (SmartBridge), "Country/region" (Indonesia), and "Work email" (silvisaumiansa@gmail.c...). At the bottom of the form, there is a checkbox labeled "I agree to the Main Services" which is checked. The browser's address bar shows the URL "salesforce.com/p".

## Step 2: Activate the Account

1. Open the activation email received from Salesforce.
2. Select the Reset Password option.
3. Create a password and answer the required security question.
4. Complete the setup process and access the Salesforce environment.

## Step 3: Create the Utility Asset Custom Object

A custom object was created through Salesforce Object Manager to represent utility infrastructure assets.

### Configuration Details

- Object Label: Utility Asset
- Plural Label: Utility Assets
- API Name: Utility\_Asset\_\_c
- Record Name: Asset Name
- Record Type: Auto Number
- Display Format: MET-{-0000}
- Starting Number: 1

The screenshot shows the Salesforce 'New Custom Object' configuration page in a web browser. The browser tabs include 'Salesforce CRM Transform', 'Skill Wallet - AI-Powered', 'M4 Data Model Design', 'Lightning Experience | Salesforce', and 'New Custom Object | Salesforce'. The address bar shows the URL: 'orgfarm-b3999ee83e-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/new'. The page header includes 'Developer Edition' and a search bar. The main content area is titled 'New Custom Object' and contains a message: 'Permissions for this object are disabled for all profiles by default. You can enable object permissions in permission sets or by editing custom profiles. Tell me more! Don't show this message again'. Below this is the 'Custom Object Definition Edit' section with 'Save', 'Save & New', and 'Cancel' buttons. The 'Custom Object Information' section includes fields for 'Label' (Utility Asset), 'Plural Label' (Utility Assets), 'Object Name' (Utility\_Asset), and 'Description'. Examples for 'Account' and 'Accounts' are provided for the label and plural label fields. A 'Starts with vowel sound' checkbox is also present. The bottom of the page shows a Windows taskbar with a weather widget (34°C, Sebagian cerah) and the date/time (16:11, 01/06/2026).

Setup  
New Custom Object

Help for this Page

Permissions for this object are disabled for all profiles by default. You can enable object permissions in permission sets or by editing custom profiles. [Tell me more!](#) [Don't show this message again](#)

Custom Object Definition Edit

Save Save & New Cancel

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label  Example: Account

Plural Label  Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name  Example: Account

Description

Additional settings enabled:

- Allow Reports
- Track Field History
- Allow Search

#### **Step 4: Configure Utility Asset Fields**

The following custom fields were added to support operational requirements:

##### **Meter ID**

Configured as a Text field with a maximum length of 50 characters. The field is designated as both Unique and External ID to support rapid search performance and prevent duplicate entries.

##### **Asset Status**

Implemented as a Picklist field containing the values:

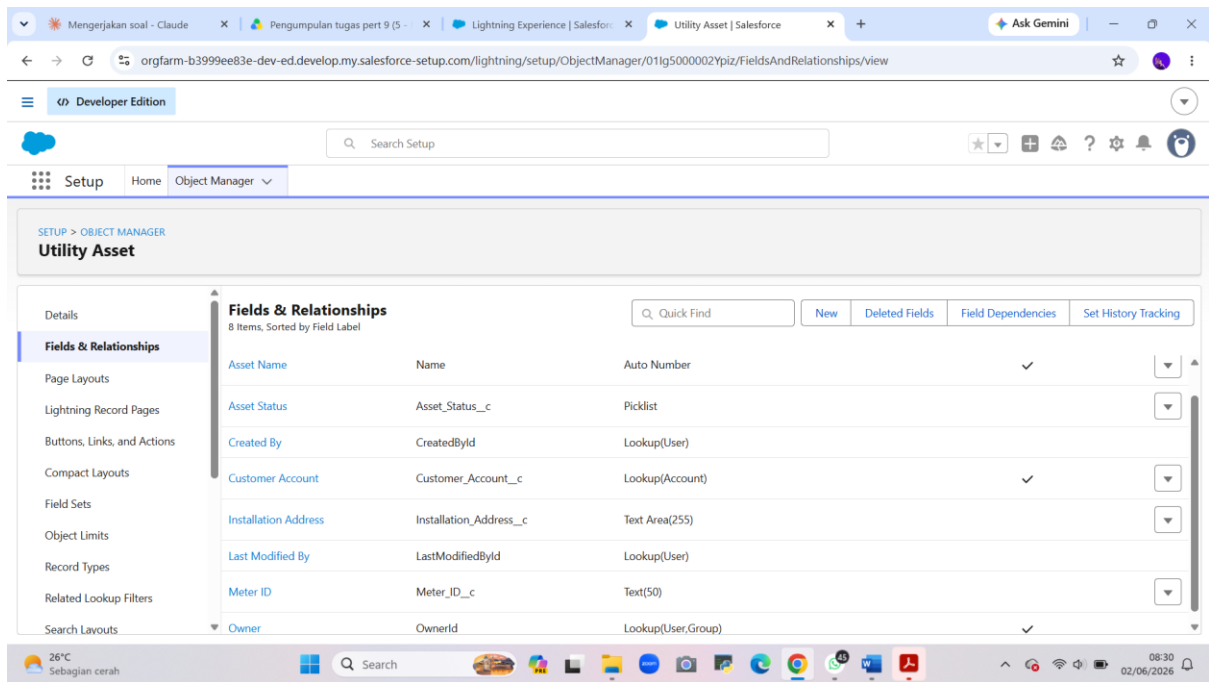
- Active
- Inactive
- Suspended

##### **Installation Address**

Configured as a Text Area field with a maximum length of 255 characters to store location information.

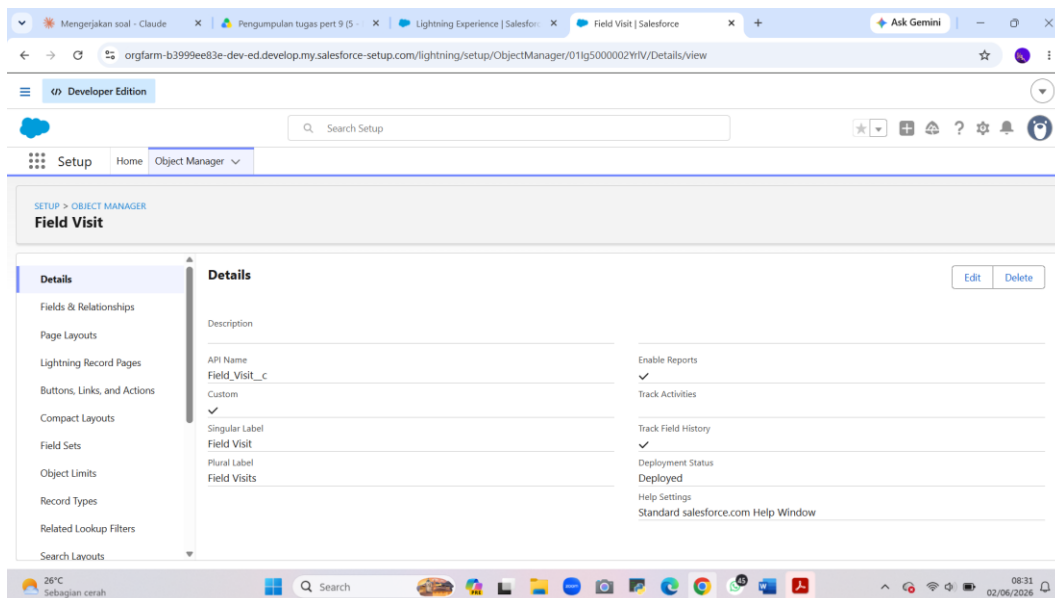
##### **Customer Account**

Implemented as a Lookup Relationship connected to the standard Account object.



## Step 5: Create the Field Visit Custom Object

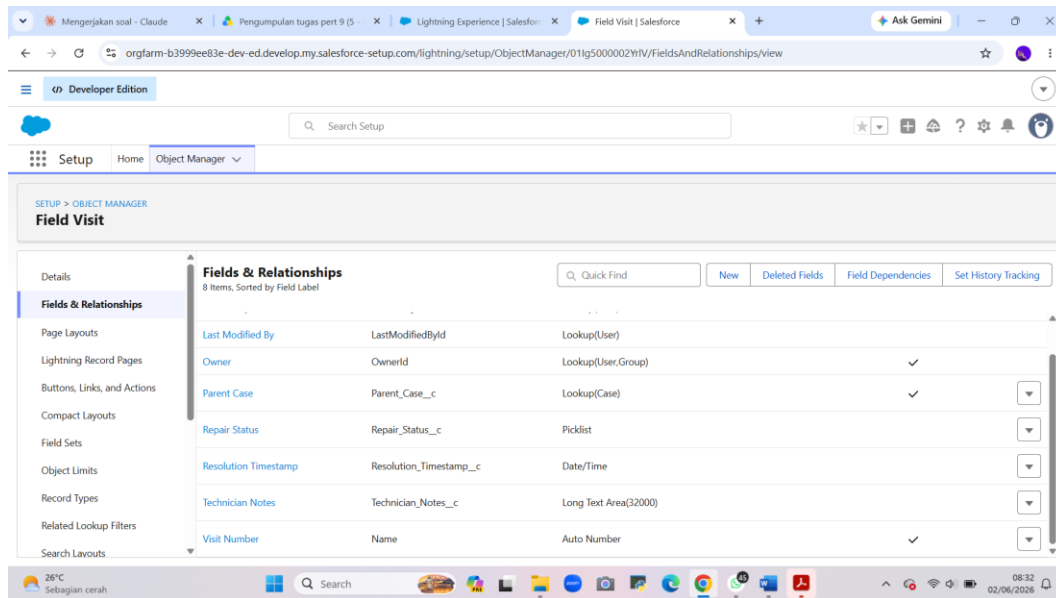
A second custom object was introduced to support technician dispatching, field operations tracking, and service completion reporting. This object replaces manual paper-based tracking processes and centralizes field activity management.





## Step 6: Configure Field Visit Fields and Relationships

Additional fields and relationship settings were created within the Field Visit object to support operational flexibility, mobile accessibility, and accurate service tracking.

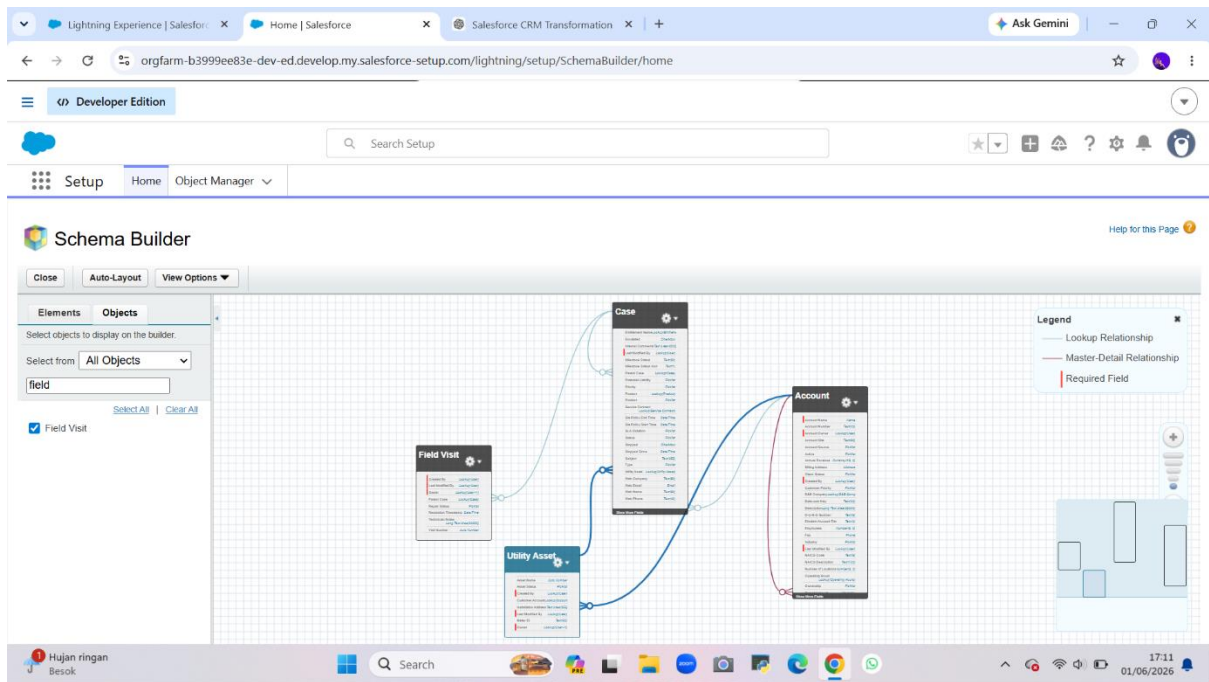


## Step 7: Validate the Data Model Structure

Salesforce Schema Builder was used to review and validate the overall database structure. Standard objects such as Account and Case, together with the custom Utility Asset and Field Visit objects, were examined to verify relationship integrity before proceeding to security configuration activities.

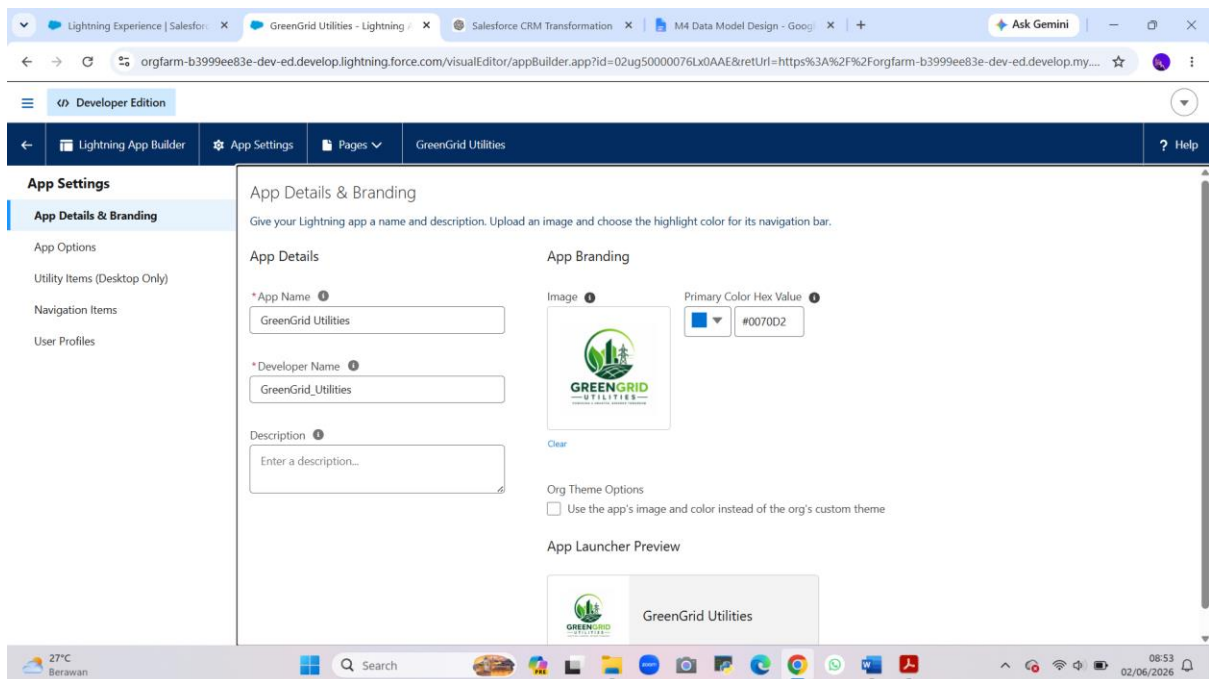
### Schema Builder Verification Steps

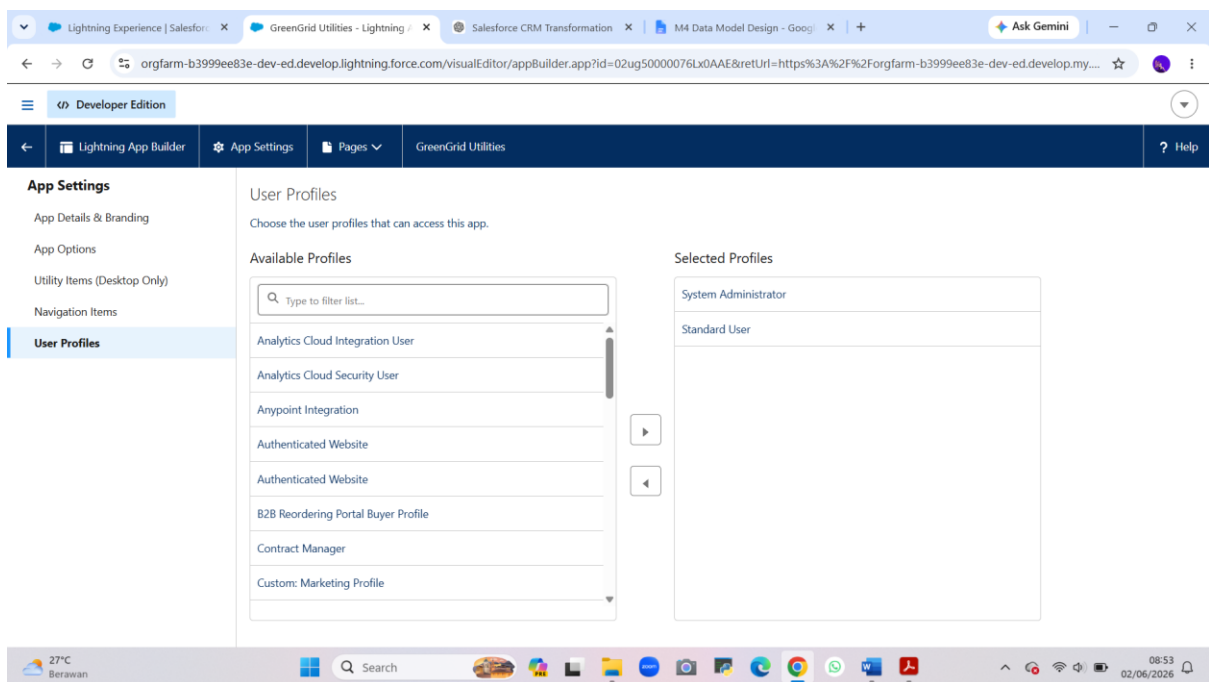
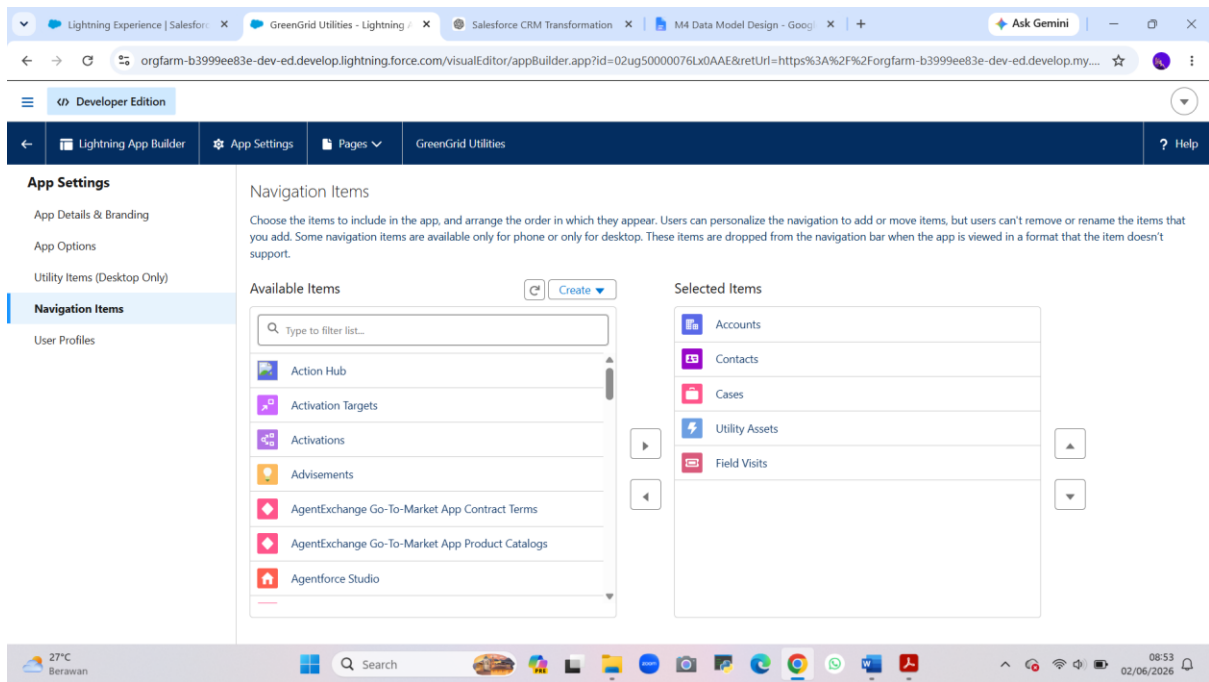
1. Open **Schema Builder** from Salesforce Setup.
2. Clear all selected objects.
3. Select the following objects:
  - Account
  - Case
  - Utility Asset
  - Field Visit
4. Review the relationship structure and verify that all connections are correctly configured.



## Step 8: Custom Lightning App Creation

To establish a dedicated, branded workspace layout for end-users, a custom Lightning Application container was provisioned within the platform App Manager.



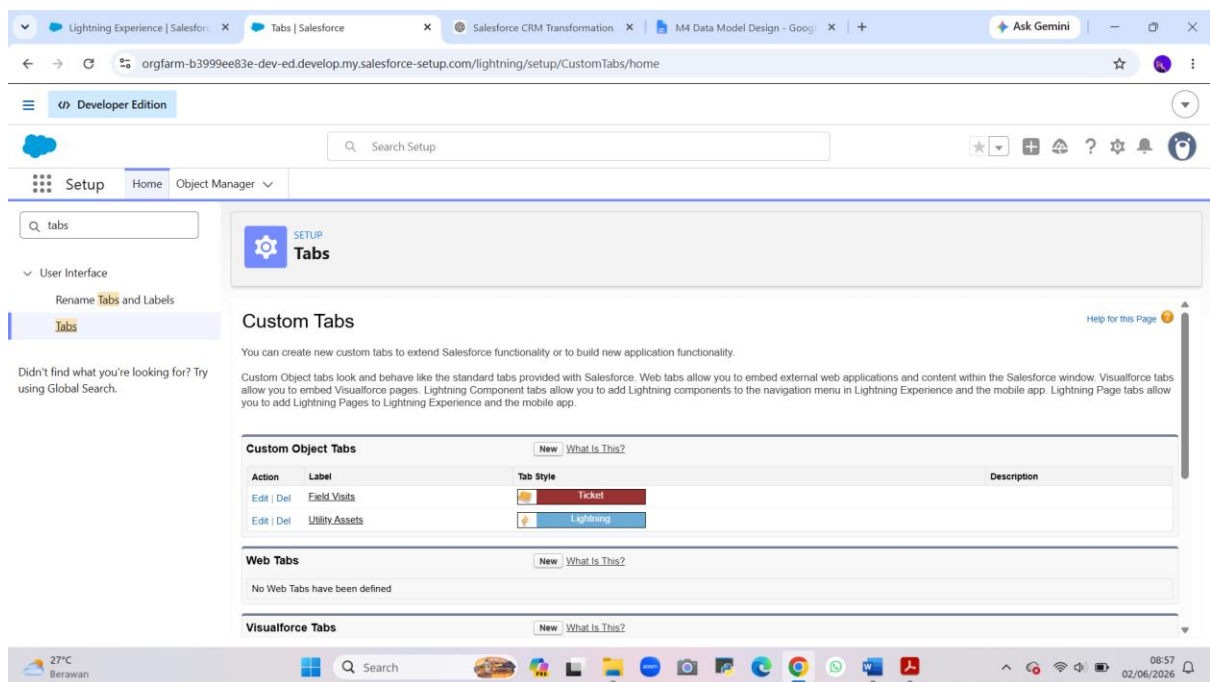


## Step 9: Custom Object Tab Provisioning (Utility Asset & Field Visit)

To make the newly deployed custom tables visible and accessible to system end-users within the navigation interface, standalone user interface tabs were provisioned.

- Navigate to **Setup** → type **Tabs** in the Quick Find bar → click on **Tabs**.
- Under the **Custom Object Tabs** related list, click the **New** button.
- Click **Next**, leave the profile visibility settings at their default values, and click **Next** again.

- On the **Add to Custom Apps** step, uncheck the "Include Tab" global header box to remove it from irrelevant background systems, but ensure the checkbox remains explicitly checked for the **GreenGrid Utilities App**.
- Ensure the checkbox for "**Append tab to users' existing personal customizations**" is checked at the bottom of the page layout.
- Click **Save**.



## Conclusion

This milestone establishes a robust Salesforce data model that supports customer management, service operations, and field maintenance activities. By combining standard and custom objects with well-defined relationships, the proposed architecture promotes data consistency, operational efficiency, and scalability. The completed data model also serves as the foundation for future automation, security configuration, and reporting initiatives within the Salesforce CRM implementation project.